

Empirical Proof Record: VALIDATED

2026-05-22T18:22:29.635644+00:00

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Proof ID: f6af1a5c15df3c4aa0f94411f77485fe...

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1. Claim

After the same documents in a different ORDER, the same identically-recognised query gets a different prime-address from Kimera (hysteresis) beyond its own run-noise, whereas a set-based retriever returns the identical ranking (order-blind by construction): $\text{order_hysteresis} = \text{mean}(\text{order_divergence}) - \text{mean}(\text{noise_floor}) > 0$.

- **Operationalisation:** Run 6 documents + 6 held-out probes through Kimera's entity target (batch) in order A, order A again (determinism control), and order B (shuffled). Per probe: $\text{order_divergence} = 1 - \text{Jaccard}(\text{prime_chain}|A, \text{prime_chain}|B)$; $\text{noise_floor} = 1 - \text{Jaccard}(\text{prime_chain}|A1, \text{prime_chain}|A2)$. $\text{order_hysteresis} = \text{mean}(\text{order_divergence}) - \text{mean}(\text{noise_floor})$. Baseline: TF-IDF retrieval order-divergence over the same docs/queries ($= 0$). VALIDATED iff $\text{order_hysteresis} > 0$ with a significant paired $\text{order_divergence} > \text{noise_floor}$ test.
- **Threshold:** $\text{order_hysteresis} > 0.0 \text{ jaccard_distance}$
- **H0:** H0: $\text{order_hysteresis} \leq 0$ — Kimera's response does not depend on document order beyond run-noise; accumulation is commutative, like a set-based retriever
- **H1:** H1: $\text{order_hysteresis} > 0$ — the same query after the same documents in a different order gets a different prime-address; Kimera carries the ORDER of experience (hysteresis) where RAG carries only the inventory

2. Verdict

VALIDATED — observed 0.9333333333333333
order_hysteresis +0.9333 = Kimera order-divergence 1.0000 – noise-floor 0.0667
over 6 probes (prime-address); RAG (TF-IDF) order-divergence 0.0000 (set-
based, order-blind by construction); paired order>noise Wilcoxon p=0.016 (sig-
nificant — order matters); manifold-state order-effect coupling Δ =0.058803,
mass Δ =4.511444 (noise coupling Δ =0.175958)

3. Evidence

Pillar	Statistic		Value	95% CI	Library
0.comparison.order_hysteresis	order_hysteresis	0.9333333333333333	—	ophamin 0.10	

4. Pre-registration

- Registered at: 2026-05-22T18:22:29.262800+00:00
- Config hash: e9e0949b499b8b4758262350a7ec8186...
- Data hash: 3896f14d8eaa26af9e1b43a0bcd7ece7...

5. Signature

37f93c4d5b33ae0e9f33f7a9be7c9d3eb0bd1a897ea3c458938d4d0bb716b809